

FORMAL LANGUAGES AND AUTOMATA THEORY

QUIZ # 4 – SOLUTION

REG#: _____

NAME: _____

COURSE CODE: CS224

INSTRUCTOR: MUHAMMAD SAJID ALI

TOTAL MARKS: 10

Question # 1: Construct the Context Free Grammar (CFG) for each of the following languages (CLO3) (10 Marks)

1. Language $L = a^n b^{2n}$ where $n \geq 0$

1. $S \rightarrow a S b b$
2. $S \rightarrow a b b$
3. $S \rightarrow \Lambda$

2. Language $L = \{ a^n b^m c^n \mid m, n \geq 0 \}$

1. $S \rightarrow a S c$
2. $S \rightarrow B$
3. $S \rightarrow \Lambda$
4. $B \rightarrow b B$
5. $B \rightarrow \Lambda$